Table of Contents

1.2.1 | Load Combinations and Geometry

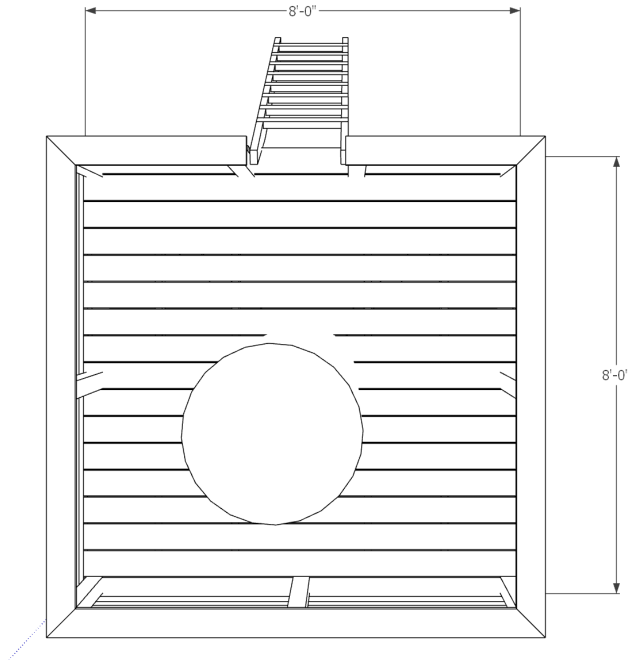

Fig. 1 - Tree Fort Plan

Table 1: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

1.2.2 | Unit Loads

Unit weights imported from csv file created by AI.

Table 2: Unit Weights - Doug Fir

Nominal	ActualWidth_in	ActualDepth_in	Area_in2	Volume_ft3_per_ft	Density_lb_per_ft3	Weight_lb_per_ft
2x2	1.5	1.5	2.25	0.015625	34	0.53125
2x4	1.5	3.5	5.25	0.0364583	34	1.23958
2x6	1.5	5.5	8.25	0.0572917	34	1.94875
2x8	1.5	7.25	10.875	0.0755208	34	2.56771

Nominal	ActualWidth_in	ActualDepth_in	Area_in2	Volume_ft3_per_ft	Density_lb_per_ft3	Weight_lb_per_ft
2x10	1.5	9.25	13.875	0.0963542	34	3.275
2x12	1.5	11.25	16.875	0.117188	34	3.98438
4x4	3.5	3.5	12.25	0.0850694	34	2.89236

Variables assigned by inline definitions.

Table 3: Member Nominal Loads and Properties

variable	value	[value]	description
D_1	2.00 plf	0.00 klf	2x6 planks DL
D_2	2.60 plf	0.00 klf	2x8 joists DL
D_3	2.90 plf	0.00 klf	4x4 posts and struts
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
LL_1	40.00 psf	1.92 kPA	ASCE7-05 floor LL
HL_1	20.00 psf	0.96 kPA	ASCE7-05 HL